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Adjustment system for mounting and fixing supports on tool bars of agricultural implements and agricultural implement

Abstract

An adjustment system for mounting and fixing a support on a toolbar of an agricultural implement includes an upper mounting assembly having an upper support block. The upper support block is configured to couple to the support via first fasteners, and the upper support block is configured to engage an upper mounting guide of the toolbar. The adjustment system also includes a lower mounting assembly having a lower support block. The lower support block is configured to couple to the support via second fasteners, and the lower support block is configured to engage a lower mounting guide of the toolbar. Furthermore, the adjustment system includes one or more eccentric bushings, and each eccentric bushing has a passing hole displaced from a center of the eccentric bushing. At least one support block has at least one opening, and a respective eccentric bushing is disposed within each opening.

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Background/Summary**CROSS-REFERENCE TO RELATED APPLICATION**

(1) This application claims priority from and the benefit of BR application Ser. No. 1020210151854, entitled "ADJUSTMENT SYSTEM FOR MOUNTING AND FIXING SUPPORTS ON TOOL BARS OF AGRICULTURAL IMPLEMENTS AND AGRICULTURAL IMPLEMENT", filed Jul. 31, 2021, which is hereby incorporated by reference in its entirety.

FIELD OF INVENTION

(2) The present invention relates, in general, to a new adjustment system for mounting and fixing supports on toolbars of agricultural implements, such as planters and seeders. More particularly, this adjustment system comprises technical, constructive and functional features designed and developed to eliminate gaps realized during the assembly and fixation of the supports of the planting row units on the toolbar of the agricultural implements. Also, the present invention deals with a new agricultural implement that has this adjustment system for mounting and fixing the supports of the planting row units on the toolbar.

BACKGROUND OF THE INVENTION

(3) Several models of agricultural implements are known in the state of the art, for example planters and seeders that are used for the distribution of seeds, fertilizers, herbicides, pesticides and fungicides on the planting rows in the field. As must be appreciated by a person skilled in the art, these models of agricultural implements can be self-propelled or not, and in the latter case, they are towed by a work vehicle, for example, a tractor.

(4) Usually, these implements have a rigid structural chassis that is mounted on wheels installed at strategic points to allow their displacement, and this structural chassis also supports, a plurality of planting row units which are assembled and installed longitudinally along the entire length of said structural frame, with each planting row unit responsible for cutting the straw, opening a furrow in the ground, applying manure/fertilizers, depositing the seeds at a desired depth, covering the furrow and compacting the ground. Normally, these implements also comprise silos and tanks intended for the storage of seeds, fertilizers, herbicides, pesticides and fungicides, which end up being supported by the aforementioned structural chassis.

(5) As must be appreciated by a person skilled in the art, these planting row units are normally installed on the toolbar through supports that are mounted in a way that they can be moved laterally to allow the definition of the distance between the planting row units and, consequently, the distance of the planting rows in the field, since, for each type of planted crop, a predetermined distance is required.

(6) In this context, it is observed that these supports for the planting row units need to include means that facilitate this spacing and the positioning between the planting row units, because throughout the year, the same agricultural implement can be used to plant and sow different crops and, therefore, the farmer needs to make these adaptations with some frequency. Therefore, for these reasons, there are known technical solutions that try to facilitate this work, for example, in some models the toolbars have rails or guides where it is possible to mount the supports and, somehow, slide them before carrying out the effective locking by means of fasteners.

(7) Despite facilitating the assembly and fixing of the planting row units, there were some inconveniences related to small gaps between the assembly components, especially between the supports and the toolbar that, during use, due to the conditions of the ground that these equipment's circulate, end up causing high levels of trepidation that can interfere in working conditions, in the control of the distribution of seeds and other agricultural products in the field that, in the end, can affect the productivity and efficiency of the process of deposition of materials in the field.

Moreover, depending on the level of trepidation, there is a risk that the fasteners will loosen and, with this, jeopardizing the displacement of the planting row units whose results can be disastrous and dangerous, both from the point of view of productivity and safety of the work performed.

(8) As must be appreciated by a person skilled in the art, these small gaps are being formed particularly due to the way these components are manufactured, which there are parts that do not require high precision, being usually cast and welded, which ends up making it difficult to obtain a standardization in the dimensional of the parts and, consequently, generating small variations that can cause the aforementioned gaps when assembling and fixing the parts on the agricultural implement.

(9) Thus, as can be seen, the agricultural implements, of the planter and seeder type, known in the

state of the art, lack a practical, functional, and mainly efficient solution, which is capable of eliminating the assembly and fixing gaps of the supports for the planting row units and, with this, eliminate the risk of falling productivity and accidents eventually caused by the vibrations usually caused by the gaps resulting from the manufacture and assembly of the parts. Therefore, these are, among others, the inconveniences, limitations and solutions that it is intended to solve and obtain with the development of the present invention.

DESCRIPTION OF THE INVENTION

(10) Thus, according to the above, the present invention aims to provide a new adjustment system for assembling and fixing supports on agricultural implements tool bars, particularly those intended for the distribution of seeds, manures, herbicides, pesticides and fungicides in the field, such as planters and seeders, and this new adjustment system was designed and developed to obtain a practical, functional and efficient solution to the problems, limitations and inconveniences revealed in the state of the art, as summarized above.

(11) More particularly, it is one of the objectives of the present invention to provide a new adjustment system for mounting and fixing supports on agricultural implements toolbars that comprises technical, constructive and functional features: capable of eliminating possible gaps obtained in the assembly and fixation of the support for planting row units on to the toolbar and, with that, guarantee the positioning and ensure the working conditions in the field of all the planting row units of the agricultural implement.

(12) Also, it is one of the objectives of the present invention to obtain an agricultural implement, such as a planter, seeder, fertilizer spreader or a sprayer, which comprises planting row units installed on the toolbar through supports that are mounted and fixed through said adjustment systems, as mentioned above.

(13) Thus, in order to achieve the technical and functional improvements summarized above, among others, the present invention refers to a new adjustment system for mounting and fixing supports on agricultural implements toolbars, designed to be applied to the mounting guides of the tool bar and it is formed by at least one upper mounting assembly and at least one lower mounting assembly, which comprise upper and lower support blocks that are associated with said support through fasteners, being at the same time at least one of said support blocks comprises at least one opening for accommodating an eccentric bushing whose the passing hole is offset from the center of that bushing.

(14) According to an embodiment of the present invention said support blocks are configured to move vertically.

(15) Still, according to another embodiment of the system, according to the present invention, these support blocks comprise only one opening. Alternatively, these support blocks comprise two or more openings.

(16) Also, according to a further embodiment of the present invention, said eccentric bushings are applied to the lower mounting assembly, the upper mounting assembly, or both mountings assemblies.

(17) In another embodiment of the present invention, said fasteners are of the screw type, or they can also be of the screw and nut assembly type.

(18) According to a possible embodiment of the present invention, the system can be configured to be applied to toolbars with linear type mounting guides, formed in a "C" type setting, or a combination of profiles.

(19) Still, according to another particular embodiment of the system, according to the present invention, said eccentric bushing can comprise a smooth or threaded external diameter.

(20) Additionally, as mentioned above, the present invention also deals with an agricultural implement that comprises a structural chassis supported by at least one assembly of wheels and also has toolbars on which several planting row units are installed through their respective supports, and the supports of this agricultural implement are mounted and fixed next to the toolbar

through an adjustment system as mentioned above.

(21) According to possible embodiments of the present invention, this agricultural implement can be, for example, a planter, seeder, fertilizer machine or a sprayer.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The features, advantages and functional improvements of the present invention, as indicated above, will be better understood by one skilled in the art from the following detailed description, as an exemplary mode, and not restrictive mode, of possible embodiments, and with reference to the figures attached, which:

(2) FIG. 1 shows a schematic perspective view of an agricultural implement model, such as a planter or seeder;

(3) FIG. 2 shows a partial top view of a toolbar with some planting row units mounted on the respective supports;

(4) FIG. 3 shows a perspective view of a segment of the toolbar with the support mounted and secured through an adjustment system in accordance with an embodiment of the present invention;

(5) FIG. 4 shows a sectional side view of the toolbar segment with the supports shown in FIG. 3;

(6) FIG. 5 shows a front view of the bottom mounting assembly of the adjustment system of the present invention; and

(7) FIG. 6 shows a sectional side view of another model of toolbar with the support mounted and fixed through the adjustment system of the present invention.

DESCRIPTION OF THE EMBODIMENTS OF THE INVENTION

(8) The invention will now be described with reference to its exemplary embodiments, and with reference to the attached figures. Such figures are schematic, and their dimensions and/or proportions may not correspond to reality, since they aim to describe the present invention in a didactic way. Furthermore, certain known and common constructive details may have been omitted for greater clarity and conciseness from the description that follows. Reference numbers used are repeated throughout the figures to identify identical or similar parts. The terms eventually used such as “above”, “below”, “front”, “back”, “right”, “left” etc., and its variants must be interpreted according to the guidance given in FIG. 1.

(9) Now, with reference to the attached drawings and, more specifically to FIG. 1, an agricultural implement **10** is represented used for the distribution of seeds, manures, fertilizers, herbicides, pesticides and fungicides on the planting row in the field, such as the so-called planters and seeders, which is being configured to be towed across the field by a work vehicle, for example a tractor. As will be appreciated by a person those skilled in the art, these agricultural implement models **10** are formed by a tongue bar **12** designed to, on the one side, couple to the work vehicle, and on the other side, support the structural chassis **16** which is mounted on at least one assembly of wheel **18** and also has toolbars **20** on which several planting row units **22** are installed. Said agricultural implement **10** also comprises silos or tanks **32** intended for the storage of seeds and/or manure/fertilizer and are usually positioned in the central portion of the structural chassis **16**.

(10) Each planting row unit **22**, as can be seen in FIG. 2, can be installed on the toolbar **20** through supports **24**, depending on the desired setting, mainly determined by the type of crop being sown and the spacing between the planting rows that is desired in the field, these supports **24** can comprise different extensions to thereby allow the planting row units **22** to be installed on the toolbar **20** one next to the other, without interference between them.

(11) An agricultural implement **10** as described herein may be, by way of reference, a model of planter produced by CNH Industrial, commercialized under the brands Case IH and New Holland.

(12) The terms referred herein as agricultural implement, planter, seeder and their variants are used

interchangeably to designate equipment, or a machine designed and configured to promote the distribution of seeds, manure, fertilizers, herbicides, pesticides and fungicides in the ground.

(13) Thus, in view of the context indicated above, and with reference now to the other attached schematic figures, it appears that the adjustment system for mounting and fixing supports on toolbars **20** of agricultural implements **10**, according to the present invention, is designed to be applied together with the mounting guides **20a** of the toolbar **20** and configured for the mounting and fixing of the support **24** of each planting row unit **22**, said adjustment system being formed by an upper mounting assembly **26** and a lower mounting assembly **28**, which comprise upper **30** and lower **32** support blocks that are associated with support **24** through fasteners **34**, wherein at least one of said support blocks **30**, **32** comprises opening **36** for accommodating an eccentric bushing **38** whose the passing hole **38a** is displaced from the center of said bushing **38**.

(14) Thus, as can be seen particularly in FIGS. **4** and **5**, at the time of installation of the support **24** it is possible that the dimensioning of the parts, for example of the support blocks **30**, **32**, of the mounting guides **20a** of the toolbar **20**, causes gaps (h) which, as reported above with respect to the state of the art, can generate significant problems and inconvenience during work operations. Thus, with the adjustment system of the present invention, when the technician tightens the fasteners **34**, said eccentric bushings **38** are rotated and, due to the configuration of the passing hole **38a**, the support block **30** moves vertically **30**, **32** until it touches the mounting guide **20a**, eliminating said gap (h) and, on this situation, the fasteners can be effectively locked. As will be appreciated by those skilled in the art, in practice, it is possible for technicians to pre-assemble the system and after that adjust said eccentric bushings **38** in order to eliminate the gaps and, only after this adjustment, technicians can perform tightening and torque control of fasteners **34**.

(15) Said support block **30**, **32**, according to possible embodiments of the present invention, may comprise only one opening **36** and, therefore, each support attachment point is used as one block, but optionally, this support block **30**, **32** may comprise two or more openings **36**, as shown in the attached figures.

(16) It is worth noting that, although the figures represent the application of the eccentric bushings **38** only in the lower mounting assembly **28**, as must be appreciated by those skilled in the art, these eccentric bushings **38** can also be applied only in the upper mounting assembly **26**, or in both top **26** and bottom **28** mounting assemblies.

(17) Optionally, according to possible embodiments of the present invention, said eccentric bushing **38** may comprise a smooth or threaded outer diameter and, in the latter case, the inner surface of the openings **36** has a corresponding configuration to interact with that outer diameter.

(18) Regarding the fasteners **34**, according to the present invention, they can be nut and bolt type, screws, or any other locking mechanism that provides the rotational movement of the eccentric bushings **38** at the time of fixing the support **24** on the toolbar **20**.

(19) The adjustment system, according to the present invention, can be applied to different models of toolbars **20**, and more particularly to different types of mounting guides **20a**. As shown in FIGS. **3** and **4**, the toolbar **20** comprises linear mounting guides **20a**, such as, for example, rulers which are welded to the body of the toolbar **20**. FIG. **5** shows another model of toolbar **20**, where the mounting guides **20** are formed in a "C" configuration and, in this case, said support blocks **30**, **32** can be mounted internally to the "C" profile. Alternatively, according to yet another model of toolbar **20**, not shown in the figures, it may comprise a mixed configuration, where one of the mounting assemblies **26**, **28** is mounted on a closed profile as a model with the configuration in "C", and the other mounting assembly **26**, **28** can be mounted on a linear profile, in both cases it is possible to apply the support blocks with the adjustment system of the present invention.

(20) In this way, it is observed that the new proposal for the adjustment system for mounting and fixing supports **24** on toolbars **20** of agricultural implements **10**, according to the present invention, manages to simply and efficiently eliminate possible gaps (h) from failures in the dimensioning of the parts and, therefore, guarantee the correct positioning of the planting row units **22**, without risk

of affecting the productivity of work operations and also the safety and integrity of agricultural equipment.

(21) In addition to these direct benefits provided by the present invention, this adjustment system can be easily applied, even in agricultural implements currently in use, that is, it is a practical solution that can improve the working conditions of equipment that are already in field without the need for high investments by the farmer.

(22) Additionally, as mentioned above, the present invention deals with an agricultural implement **10**, which comprises a structural chassis **16** that is supported by at least one set of wheels **18** and also has toolbars **20** on which several planting row units **22** are installed through respective supports **24**, each support **24** being mounted and fixed together with the toolbar **20** with an adjustment system as defined and presented above.

(23) According to possible embodiments of the present invention, that agricultural implement **10** can be a planter, seeder, fertilizer spreader or a sprayer.

(24) Finally, considering all of the above, it is important to note that the present description aims only to present and define, in an exemplary way, particular embodiments of the new adjustment system for mounting and fixing the supports **24** on toolbars **20** of agricultural implements **10**, as defined by the present invention. Therefore, as a person skilled in the art should appreciate, several modifications and combinations of elements and equivalent details are possible without, therefore, deviating from the scope of protection defined by the attached claims.

Claims

1. An adjustment system for mounting and fixing a support on a toolbar of an agricultural implement, comprising: at least one upper mounting assembly comprising an upper support block, wherein the upper support block is configured to couple to the support via first fasteners, and the upper support block is configured to engage an upper mounting guide of the toolbar; at least one lower mounting assembly comprising a lower support block, wherein the lower support block is configured to couple to the support via second fasteners, and the lower support block is configured to engage a lower mounting guide of the toolbar; and one or more eccentric bushings, wherein each eccentric bushing of the one or more eccentric bushings has a passing hole displaced from a center of the eccentric bushing; wherein at least one of the upper support block or the lower support block comprises at least one opening, and a respective eccentric bushing of the one or more eccentric bushings is disposed within each opening of the at least one opening.
2. The adjustment system according to claim 1, wherein the upper and lower support blocks are configured to move vertically.
3. The adjustment system according to claim 1, wherein the upper support block comprises the at least one opening, and the at least one opening of the upper support block comprises only one opening.
4. The adjustment system according to claim 1 wherein the upper support block comprises the at least one opening, and the at least one opening of the upper support block comprises two or more openings.
5. The adjustment system according to claim 1, wherein at least one fastener of the first fasteners, at least one fastener of the second fasteners, or a combination thereof, is of a screw type or a set of a screw and a nut.
6. The adjustment system according to claim 1, wherein at least one of the upper mounting guide or the lower mounting guide comprises a linear type mounting guide or a “C” type setting.
7. The adjustment system according to claim 1, wherein at least one of the one or more eccentric bushings has a smooth or threaded external surface.
8. The agricultural implement, comprising: a structural chassis supported by at least one assembly of wheels; the toolbar; and a planting row unit coupled to the toolbar via the support, wherein the

- support is mounted and fixed on the toolbar through the adjustment system as defined in claim 1.
9. The agricultural implement according to claim 8, wherein the agricultural implement is a planter, a seeder, a fertilizer spreader, or a sprayer.
10. The adjustment system according to claim 1, wherein the lower support block comprises the at least one opening, and the at least one opening of the lower support block comprises only one opening.
11. The adjustment system according to claim 1, wherein the lower support block comprises the at least one opening, and the at least one opening of the lower support block comprises two or more openings.
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